

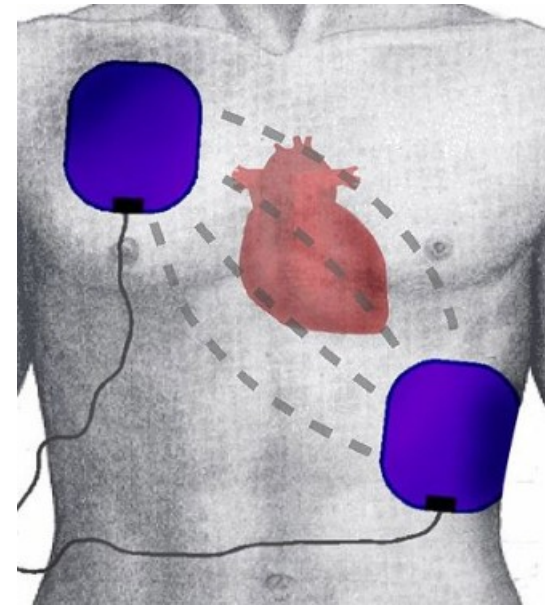
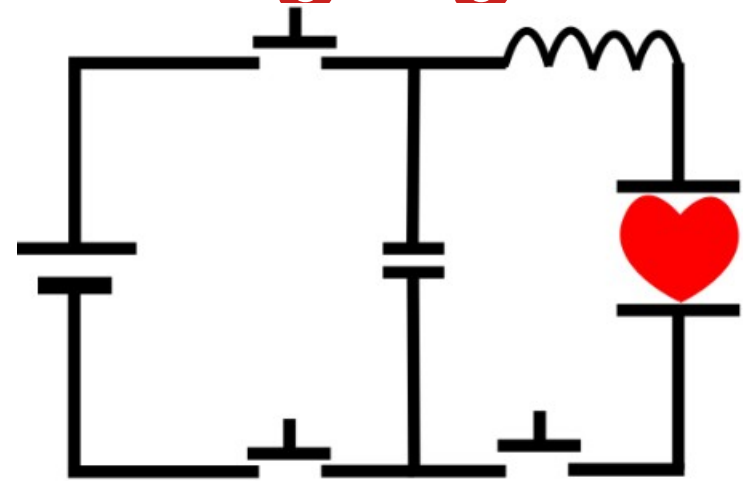
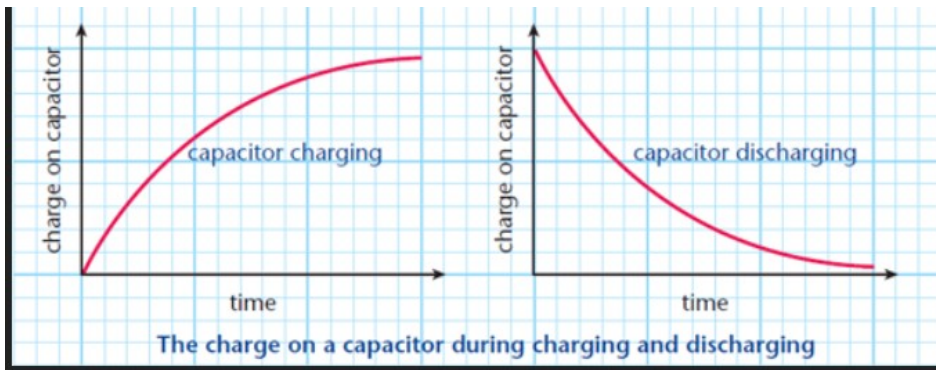
Capacitors - charging , discharging



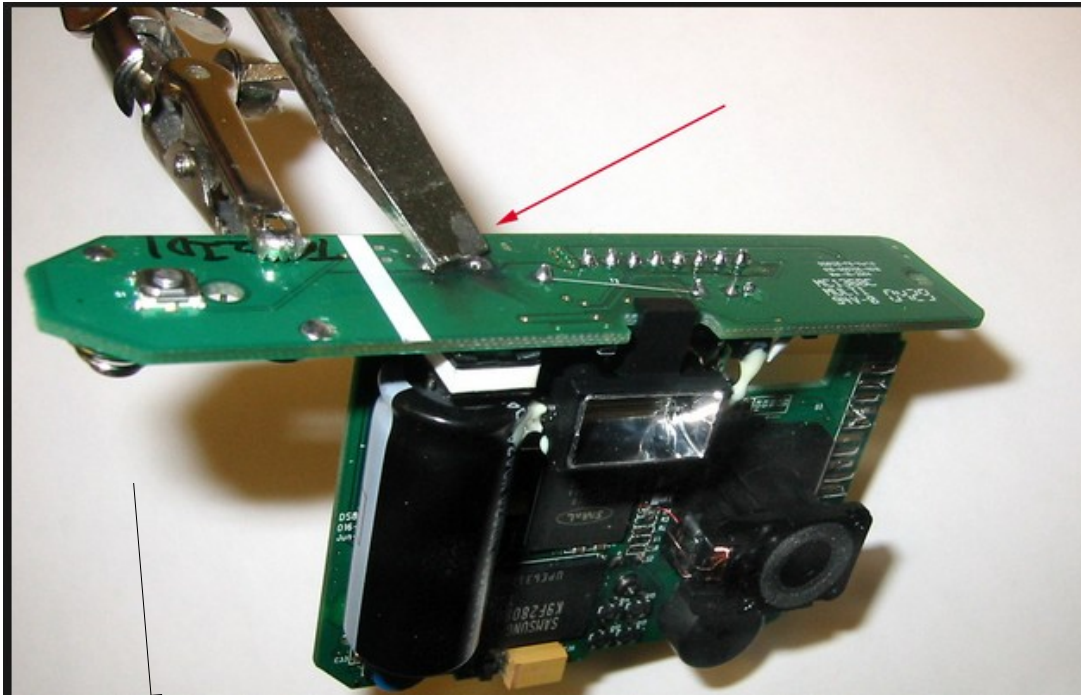
<http://micro.magnet.fsu.edu/electromag/java/capacitor/index.html>

https://phet.colorado.edu/sims/html/capacitor-lab-basics/latest/capacitor-lab-basics_en.html

demonstration



200V → 1000V for heart.



demo

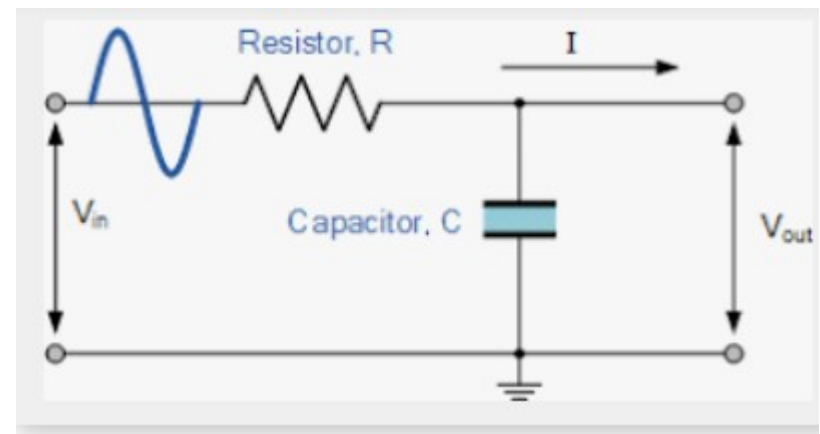
3300uF

See Harold Edgerton (30s) →
 Made high speed photography possible with
 His flash-tube He used a capacitor
 That dump the energy in a bulb that contained
 Xenon gas → plasma discharge (intense
 Burst of light in uS)
 → strobes and high speed photography

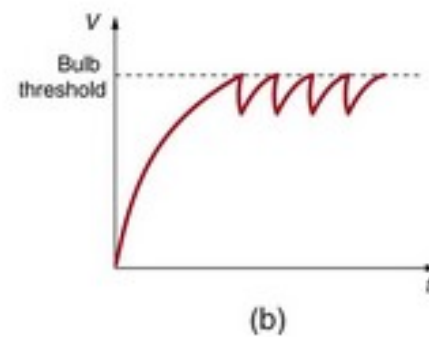
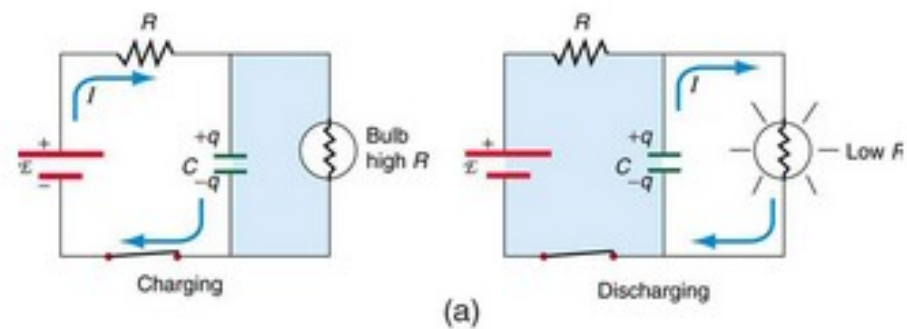
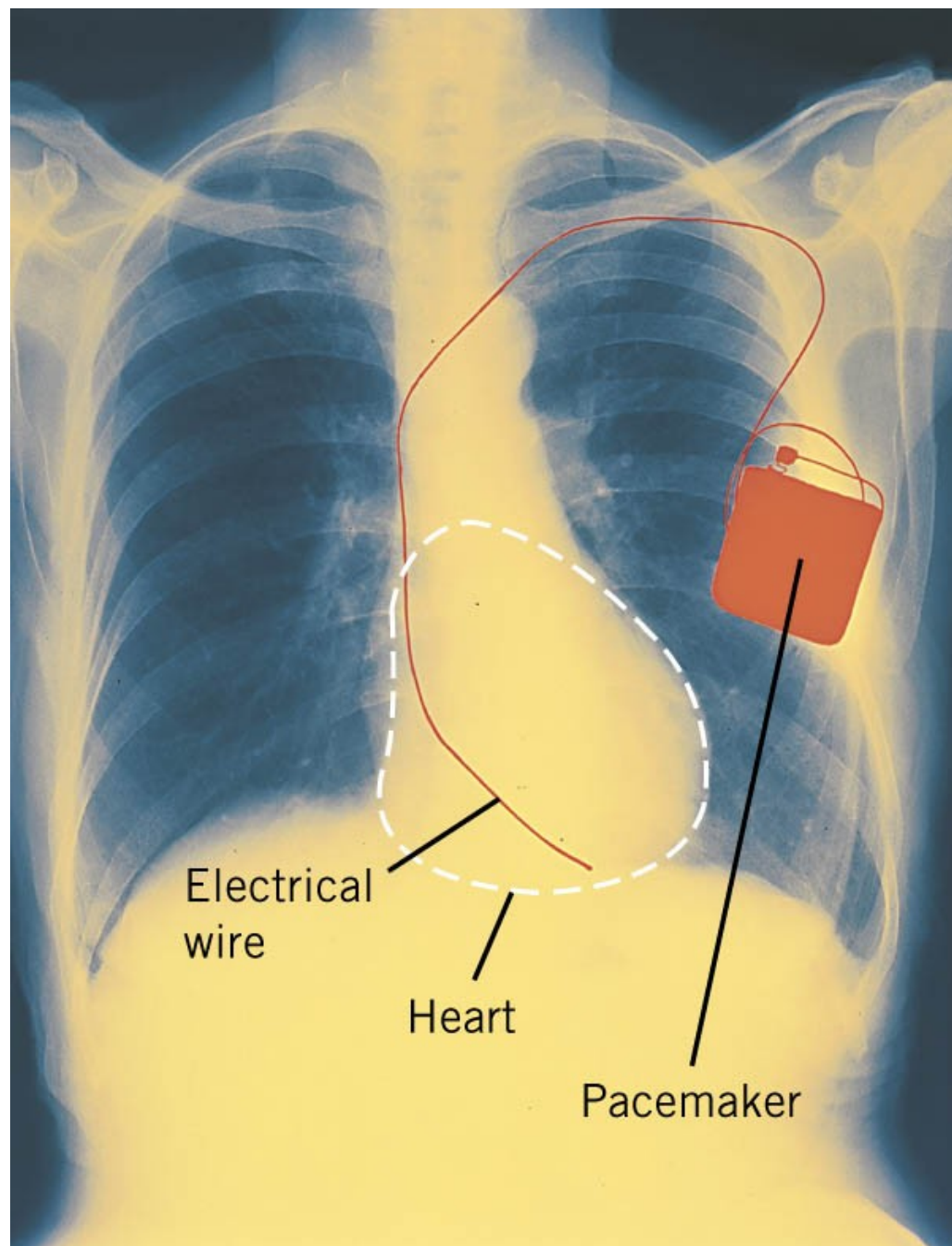
<http://www.ladyada.net/make/sudc4kap/make.html>

- Holds binary numbers (RAM memory)
- make keyboards
- filter noise
- block DC, turn AC to DC
- battery

<https://www.youtube.com/watch?v=EoWMF3Vkl6U>
 4:43 and 5:23



20.13 RC Circuits



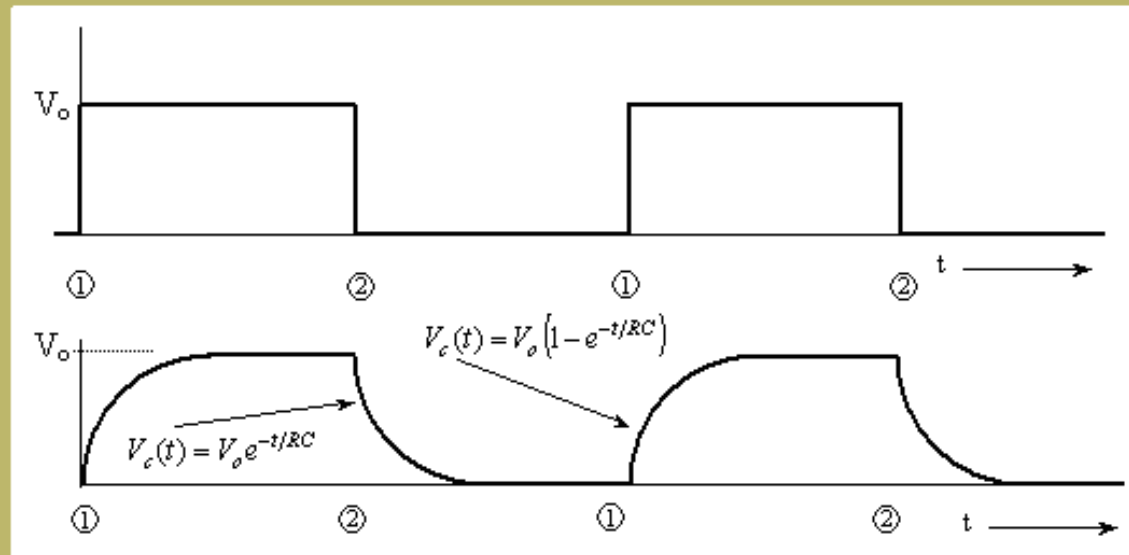
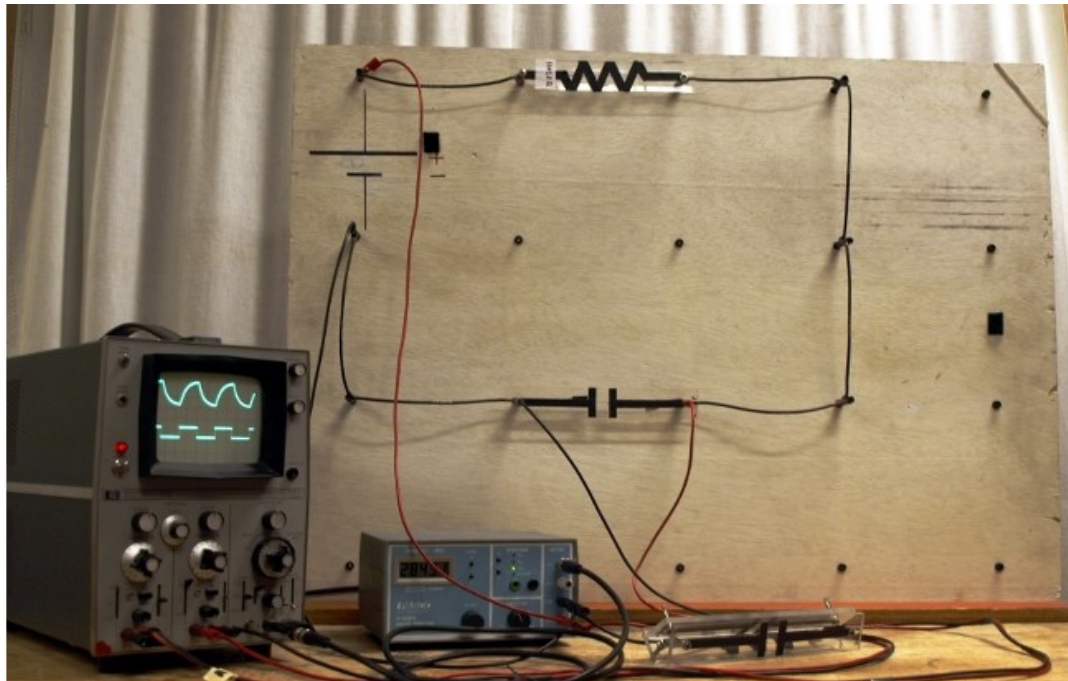
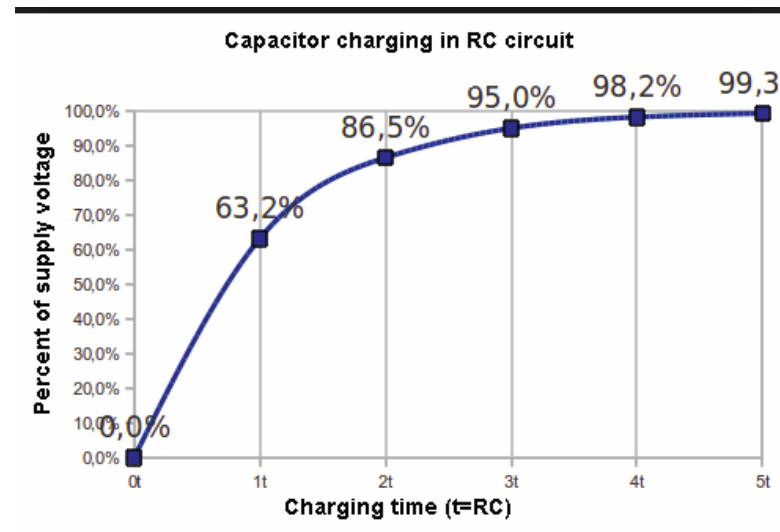
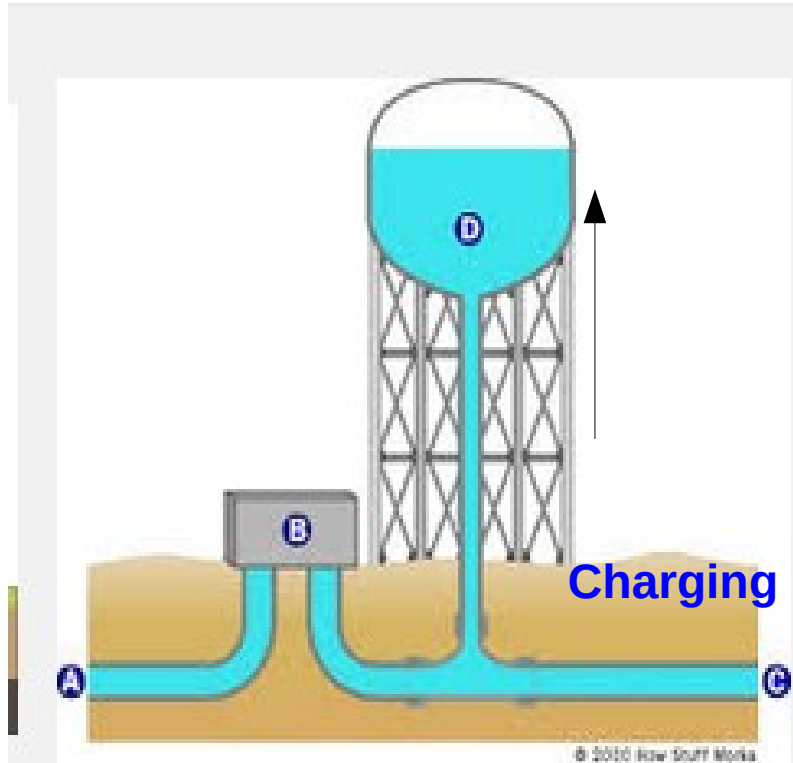


Figure 7. A square wave with a longer period drives the RC circuit and the voltage response across the capacitor is slightly different than that shown in Figure 6b. It can be assumed that the circuits are the same for Figures 6 and 7. The capacitor charges in the same amount of time as the one shown in Figure 6b, but because the square-wave period is longer, the capacitor here remains charged until the input voltage drops once again to zero.



Charging → voltage across the cap as a function of time

Note: charge follows the same graph ($Q = C V$) – the current decreases.

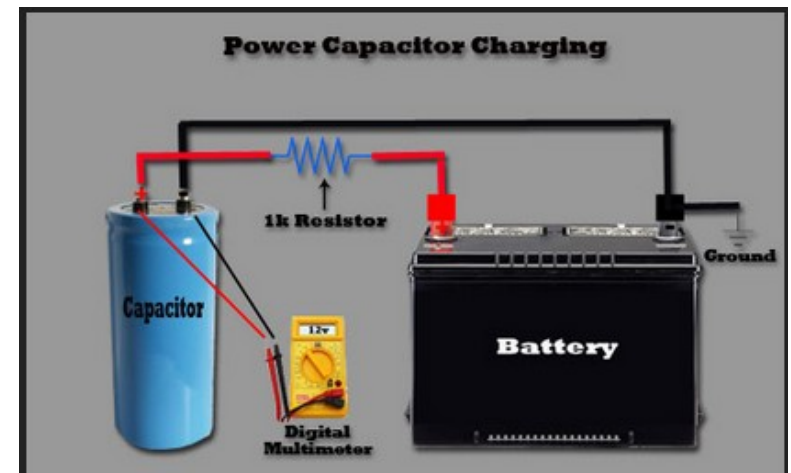


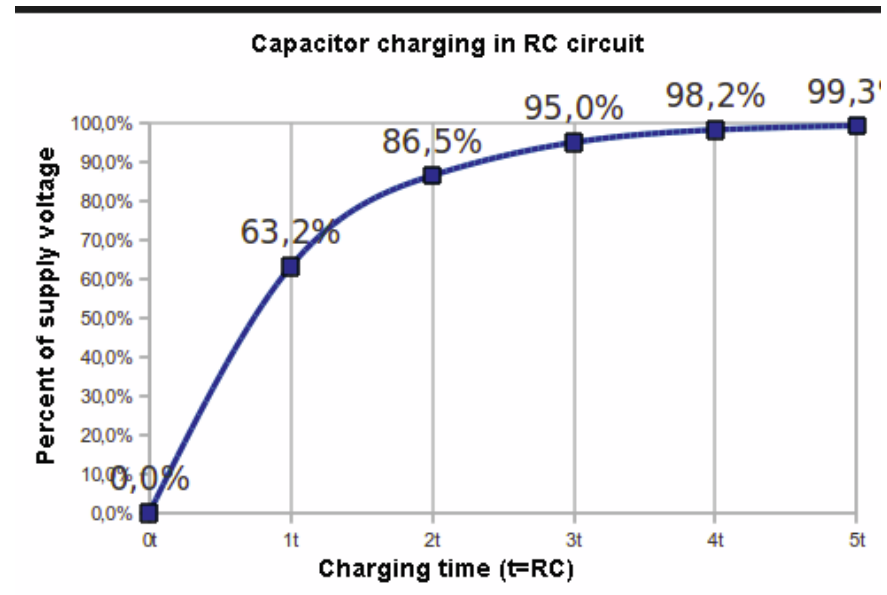
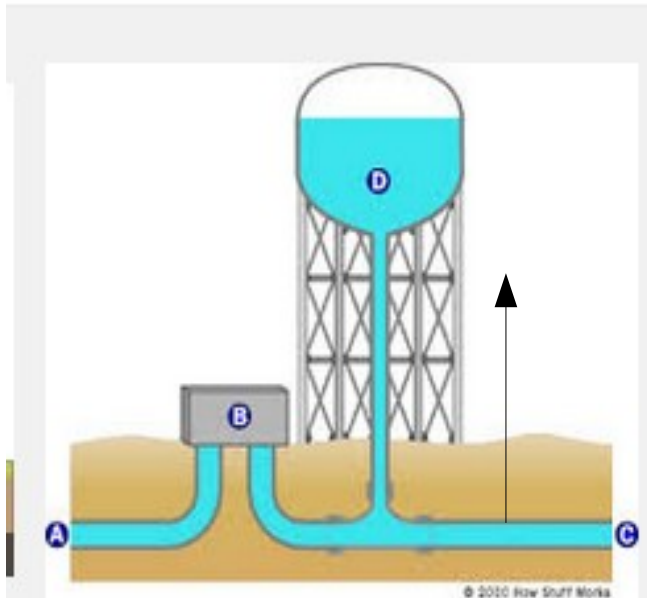
Charging → → At the beginning it is easy to pump the water up
Then it gets harder because of the weight of
The water. Same idea when you charge a capacitor

$T = RC$ is called the time constant of the capacitor.
It is the time it takes to charge it at 63%.

Do exploration of Physics app (RC)

Do some Math.





<http://micro.magnet.fsu.edu/electromag/java/capacitor/index.html>

Equation of charge **$V(t) = V_{\max} (1 - e^{-(t/RC)})$**

$V_{\max}$ is the voltage of power supply used

To charge the capacitor.

R is the resistance through which the cap is charging

C is the capacitance of the capacitor.

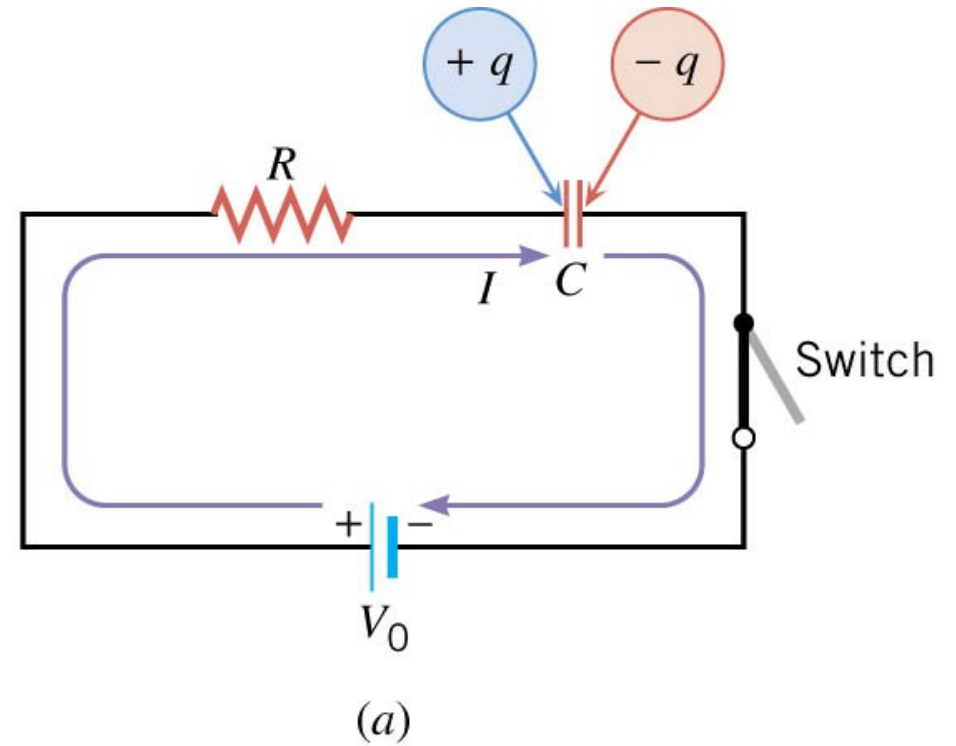
$V(t)$ is the voltage across the capacitor.

Suppose you are charging a capacitor through a resistor of 10 ohms.
The capacitance is 0.2F . What is the equation?

20.13 RC Circuits

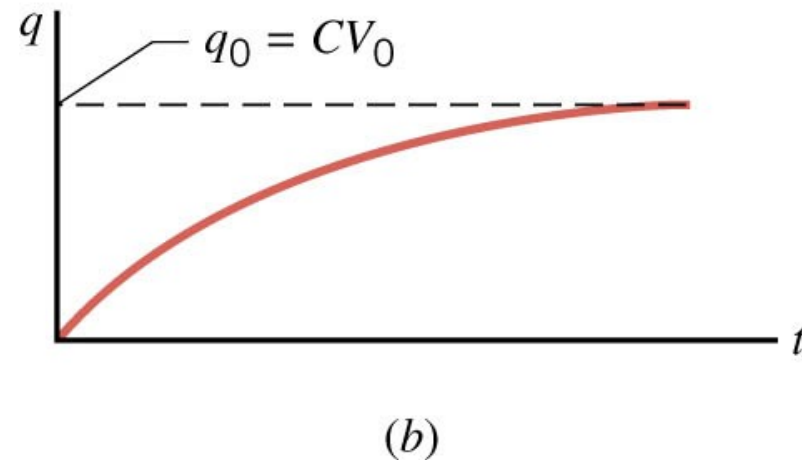
Capacitor charging

$$q = q_0 \left[1 - e^{-t/RC} \right]$$

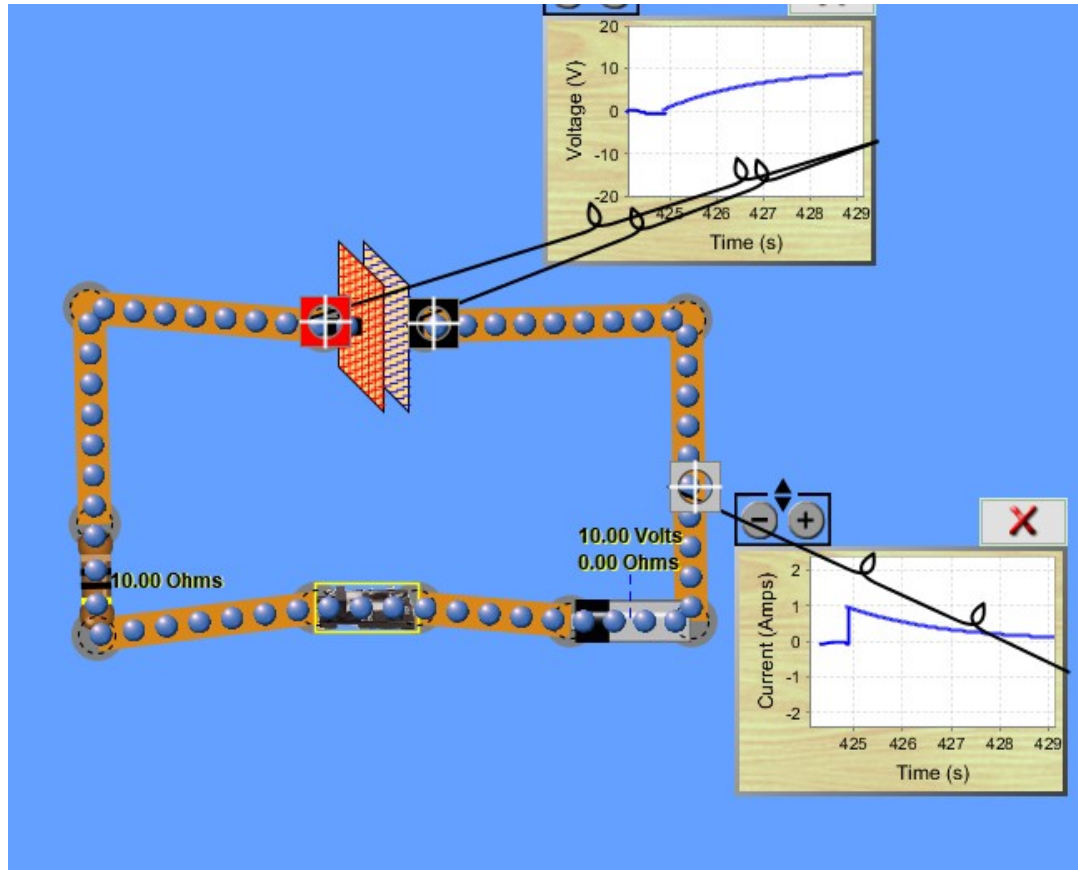


time constant

$$\tau = RC$$



Charging → voltage across the cap as a function of time



This is charging a capacitor
Through a resistance

The voltage of the battery is 10V
The resistance is 10.00 ohms
The capacitance is 0.2 Farads

The time constant is $RC = 2s$
It is the time to get to 63% of the
Maximum voltage 10V.

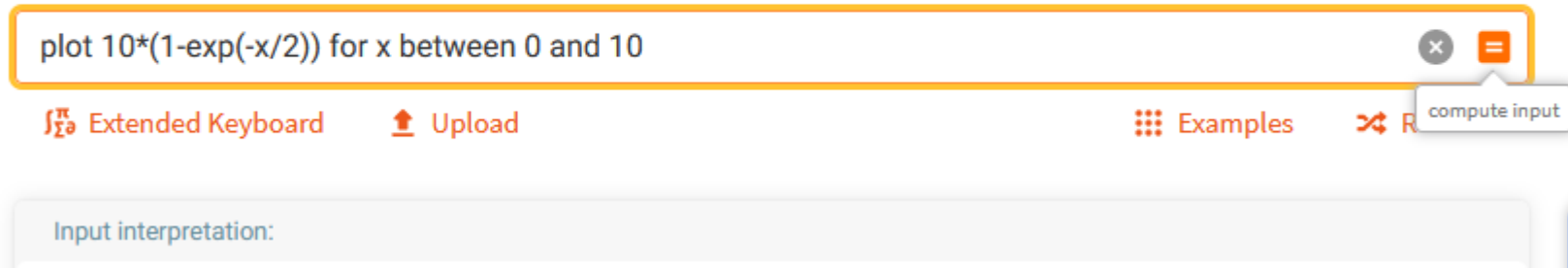
Observe the voltage across the
Capacitor. It is increasing until
It reaches 10V.

What is 63% of the max value ?
Check if after $t = RC$ the voltage is
Indeed about 63% of 50V

This graph shows the voltage across the capacitor
as a function of time. $V(t)$. The voltage increases from
0 to 10V. First the voltage increases quickly then
Slowly. The slope is steep then flats out.

Note for me: capacitor_app.jar from phys116

[https://phet.colorado.edu/en/simulation/
circuit-construction-kit-ac](https://phet.colorado.edu/en/simulation/circuit-construction-kit-ac)



1) Suppose you are charging a capacitor through a resistor of 10 ohms. The capacitance is 0.2F . What is the equation?

The equation of the charge is:

$$V(t)=10(1-e^{-t/2})$$

Plug that in wolfram alpha.

Check that when $t=2s$

The voltage is about 63% of 10

Use the equation to find the voltage when

$t=1s, 3s, 4s, 5s$ (use your TI then check with the graph

2) Suppose you are charging a capacitor with a power supply of 20V

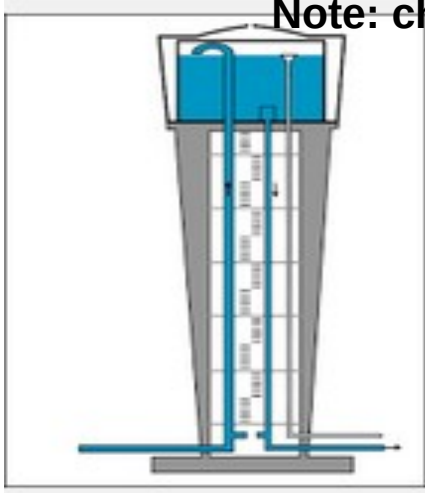
The capacitance is 0.1F and the resistance is 100 ohms

What is RC ? What is the voltage across capacitor when $t = RC$

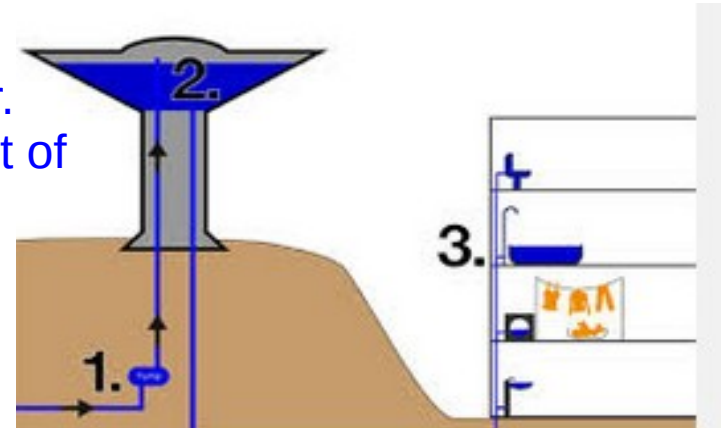
What is the equation $V(t)$? What is the voltage when $t = 5s, 10s, 20s, 30s$? Use TI)

discharging → voltage across the cap as a function of time

Note: charge follows same pattern



Like emptying the tower.
First it's fast. The weight of
The water is pushing
Then it slows down.

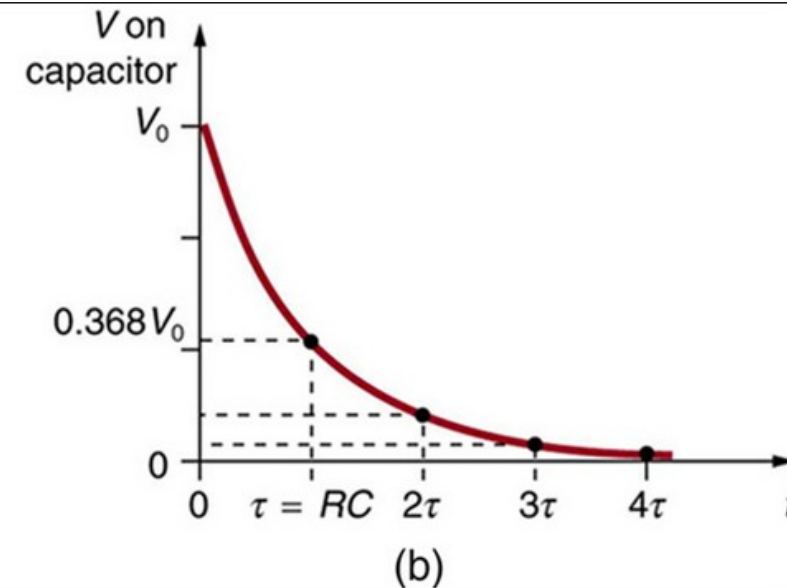
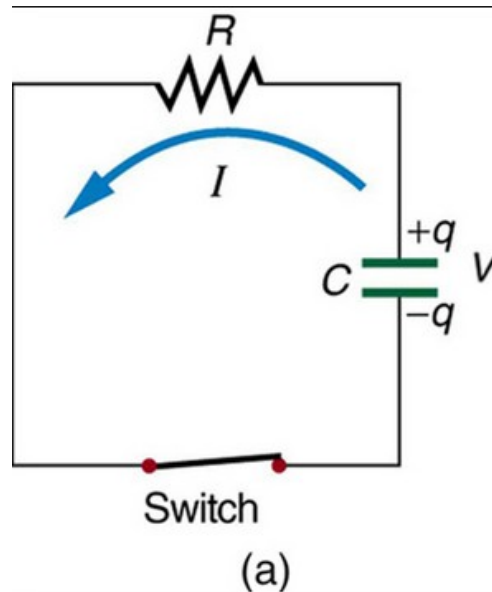


<http://micro.magnet.fsu.edu/electromag/java/capacitor/index.html>

RC is the time for the
Capacitor to lose
63% of its charge
(or voltage)
It remains 37% of the
initial voltage.

It is an exponential decay

$$V = V_{\max} e^{(-t/RC)}$$



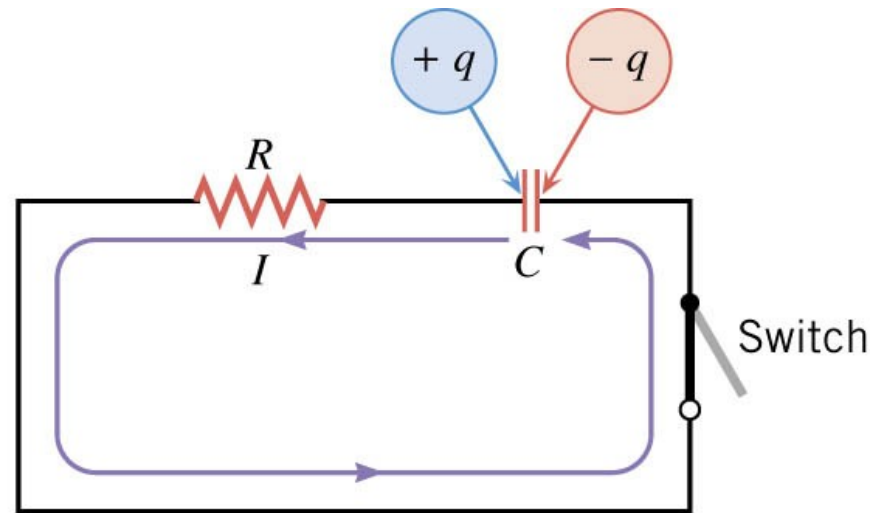
20.13 RC Circuits

Capacitor discharging

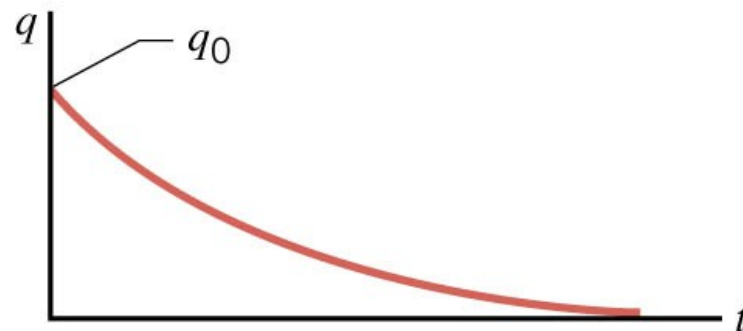
$$q = q_0 e^{-t/RC}$$

time constant

$$\tau = RC$$

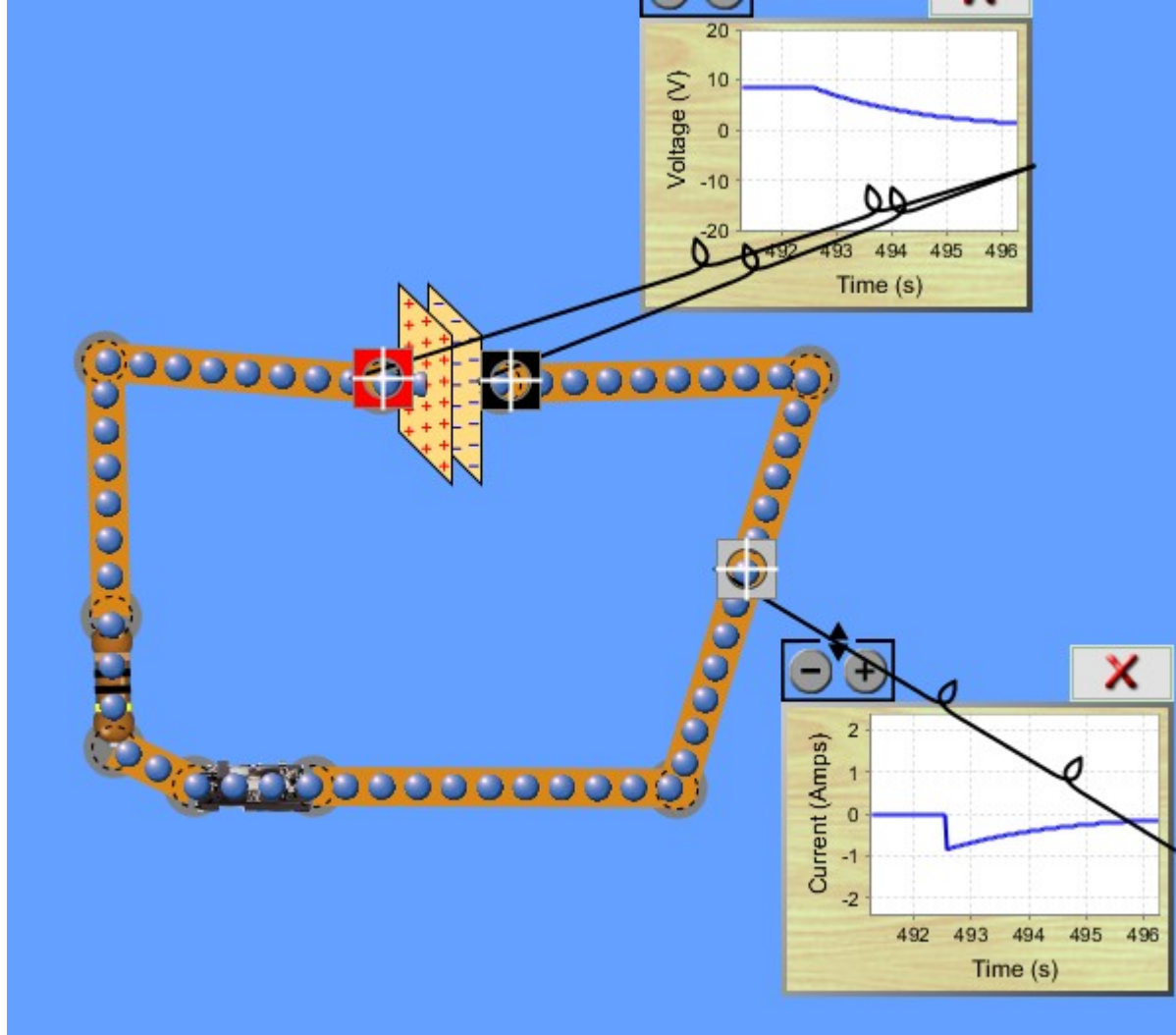


(a)



(b)

Discharging in a load



ohms

plot $10 \cdot (\exp(-x/2))$ for x between 0 and 10

Extended Keyboard Upload

Input interpretation:

plot	$10 \exp\left(-\frac{x}{2}\right)$	$x = 0 \text{ to } 10$
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Use wolfram alpha to plot the exponential

With the TI, find V for $t = 0, 1, 2, 3..$
Check with the line

A 10V voltage is used to charge a capacitor through a 500 ohms resistor.

The capacitance of the capacitor is 0.01 F (micro farad)

A) What is the time constant in seconds ?

B) The capacitor is now discharging in the resistance.

C) What is the equation of the voltage across the capacitor ?

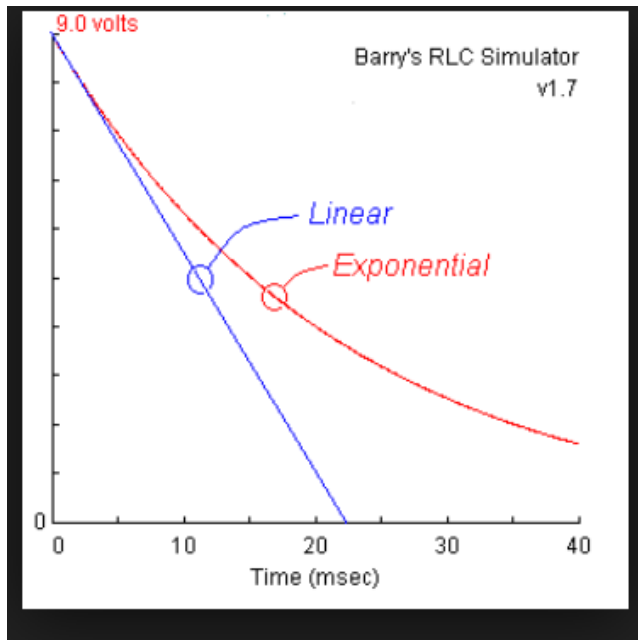
D) Plot the equation for voltage for x between 0 and 10

E) Use the equation to find the voltage at 2 s , 4 s , 6s, 8s check with graph,

F) The time constant is the time it takes for voltage to decrease by 37%. 37% of 10 is _____ -

Use your graph to find the time it takes for the voltage to decrease by 37%

Compare to the time constant $R * C$



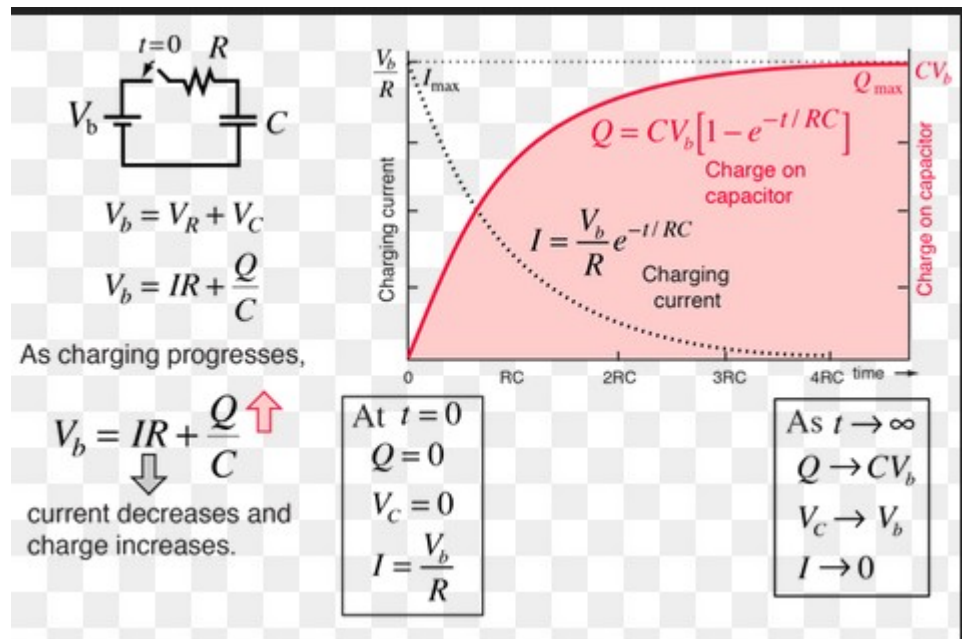
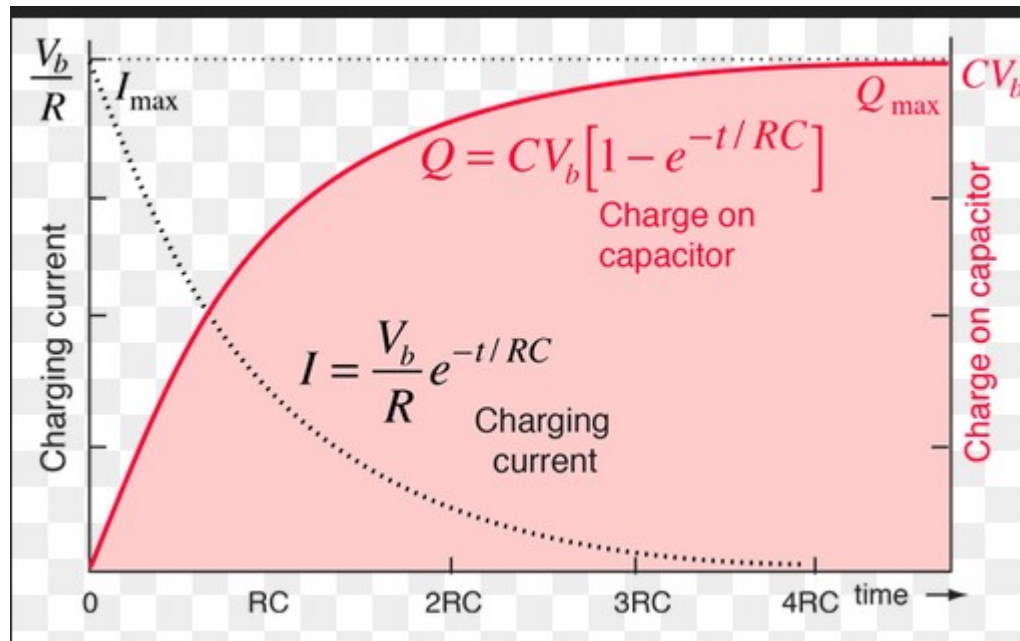
$$V = V_0 e^{-\frac{t}{RC}}$$

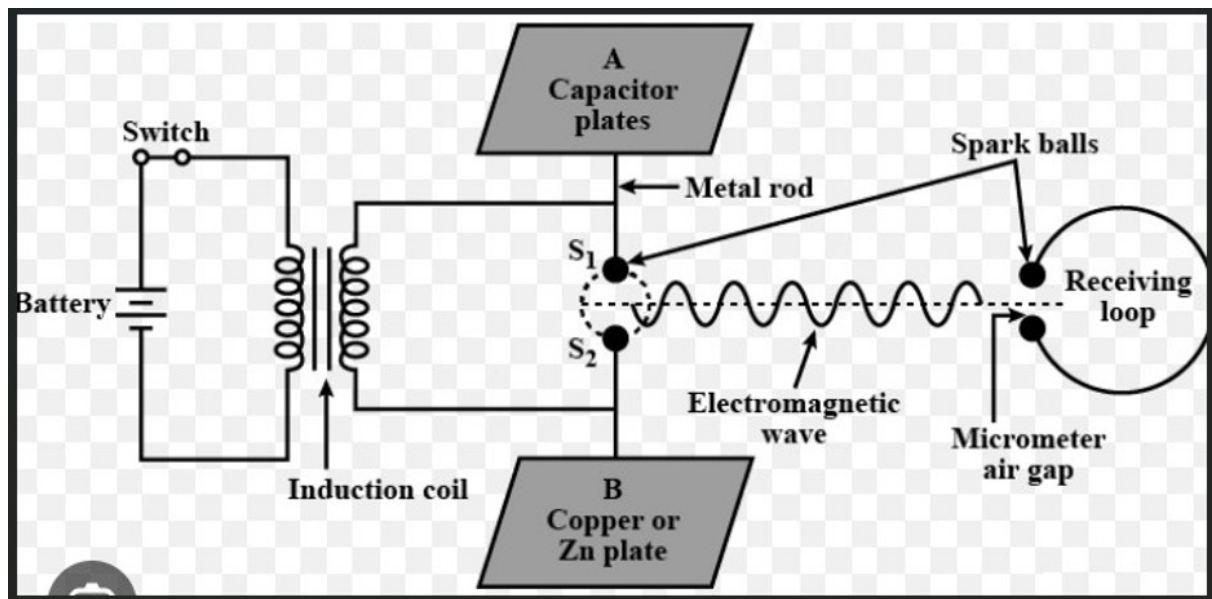
$$\ln V = \ln(V_0) - t / RC$$

1/RC is the slope of the line.

Using oscilloscope:







1887 Heinrich Hertz
Detects EM waves predicted by Maxwell.

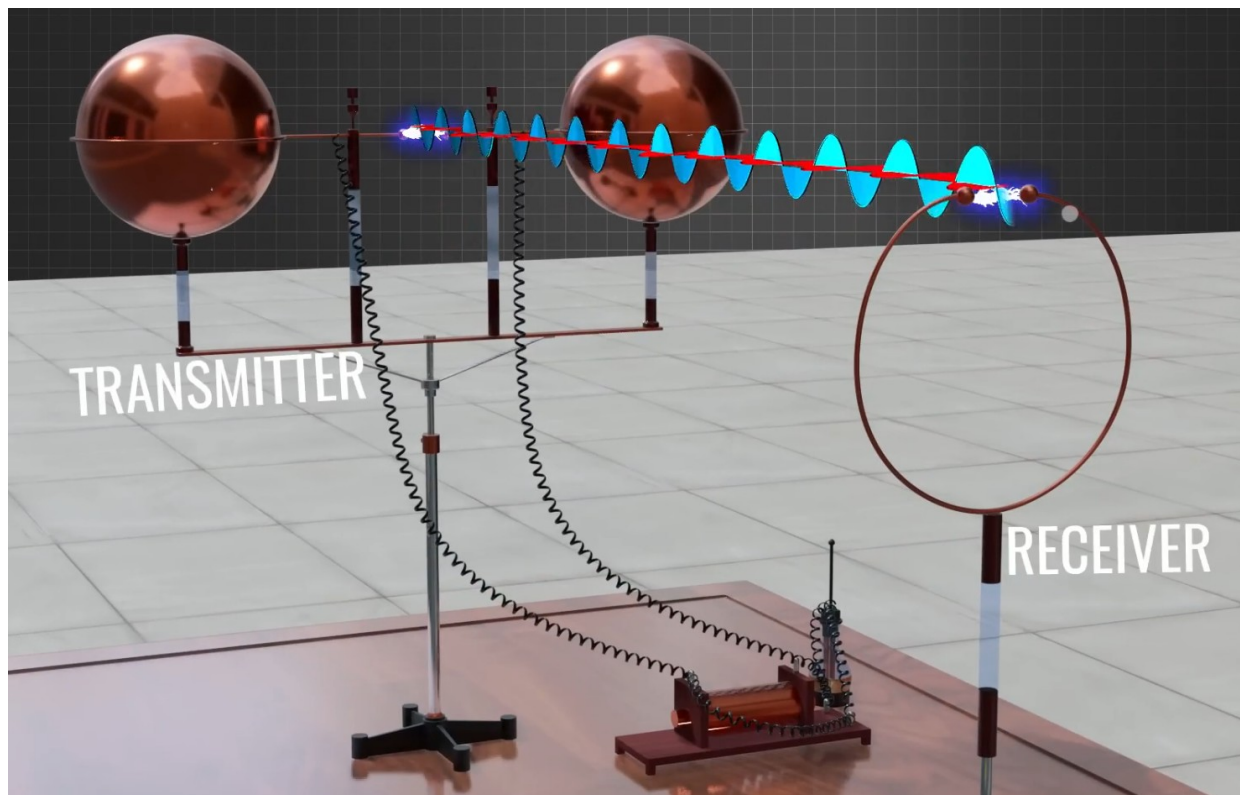
He used a RLC circuit (resonant circuit) but it was very smart. Specially because there was no AC yet.

→ he used an mechanical switch
To simulate AC to step up the voltage in the induction coil.

→ the capacitor is charging.
Current can not jump the gap.

→ voltage across the capacitor is Large enough. Breakdown of the E between the plates. There is a spark. Capacitor is discharging. Current oscillates back and forth (tank circuit. Resonance)

→ changing current create
An electromagnetic wave
Detected by the receiver.
(antenna)





1891 Nikola Tesla

Tesla coil



The capacitor is storing energy until its large enough. It is dumped at the end
In the secondary coil as a high voltage and low current. Capacitor is a way to build up energy
Like a reservoir.

Tesla coil. First the capacitor is charging (transformer). The current has nowhere to go because
Of the gap. It is until its voltage is large enough for Current to flow (oscillate) in the circuit:
capacitor, gap, primary coil. Energy is transferred to the secondary coil.
Which is a resonance cavity. The electromagnetic wave is confined inside. Reinforcing itself.
Its a way of energy transfer. The voltage is large at the top. And 0 at the bottom of the
Secondary coil.

<https://www.youtube.com/watch?v=Jl3VpKF8IF0>

https://www.youtube.com/watch?v=fT5_MmeSeDU

Schematic:

